HCIA-AI V4.0 Code-Writing & Fill-in-the-Blank Workbook

40 High-Yield Exam Code Completion Questions • Essential Hands-on Reference for PyTorch, MindSpore & Scikit-Learn

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EXAM CODE COMPLETION BLUEPRINT & SYNTAX CONVENTIONS

1. Case-Sensitivity & Capitalization
- Method & function names are strictly lowercase: zero_grad, backward, step, fit, predict
- Base classes & loss functions use PascalCase: nn.Module, nn.Cell, nn.CrossEntropyLoss

2. The Parentheses Standard
- If parentheses already exist outside the blank: Prompt: optimizer.____() → **Enter only:** zero_grad
- Do not write double parentheses like zero_grad()().
- If no () is present, both token and function are accepted.

3. Framework Core Paradigms
- **PyTorch:** Custom models implement def forward(self, x):
- **MindSpore:** Custom models implement def construct(self, x):
- **PyTorch Base:** nn.Module | **MindSpore Base:** nn.Cell

4. Scikit-Learn Trailing Underscores
- All attributes learned during fit() end with an underscore: coef_ (weights), intercept_ (bias), cluster_centers_
- **Data Preprocessing:** fit_transform() on training data, but strictly transform() on test data to avoid leakage.

Category 1: PyTorch Deep Learning Training & Workflow (Q1 – Q15)

QUESTION 1 • PyTorch Autograd — 16 Points

In the PyTorch training loop, which method must be called on the optimizer before calculating gradients to clear previously accumulated gradient values?

```
# PyTorch Training Loop Step 1:
optimizer.____()
outputs = model(inputs)
loss = criterion(outputs, labels)
loss.backward()
optimizer.step()
```

CORRECT CODE TOKEN: `zero_grad (or zero_grad())`

Technical Breakdown & Key Insight: PyTorch accumulates gradients by default across iterations. Calling optimizer.zero_grad() clears existing gradients before computing the new batch gradients.

QUESTION 2 • PyTorch Backpropagation — 16 Points

In PyTorch, which method is invoked on the computed scalar loss tensor to execute backpropagation and calculate the gradients of the loss with respect to all model parameters?

```
outputs = model(inputs)
loss = criterion(outputs, labels)

# Execute reverse-mode automatic differentiation:
loss.____()
optimizer.step()
```

CORRECT CODE TOKEN: `backward (or backward())`

Technical Breakdown & Key Insight: Calling loss.backward() computes the derivative of the loss with respect to all graph leaves (trainable parameters) using the chain rule.

QUESTION 3 • PyTorch Optimizer Update **16 Points**

In PyTorch, which method of the optimizer is called to update the model parameters (weights and biases) according to the computed gradients?

```
optimizer.zero_grad()
loss.backward()

# Update parameters using computed gradients:
optimizer.____()
```

CORRECT CODE TOKEN: `step (or step())`

Technical Breakdown & Key Insight: optimizer.step() applies the optimization algorithm formula (such as SGD or Adam) to update all weight and bias tensors in-place.

QUESTION 4 • PyTorch Model Inheritance **16 Points**

In PyTorch, all user-defined neural network architectures must inherit from which base class in the torch.nn package?

```
import torch.nn as nn

# Base class for all neural network modules:
class LeNet(nn.____):
    def __init__(self):
        super(LeNet, self).__init__()
        self.conv1 = nn.Conv2d(1, 6, 5)
```

CORRECT CODE TOKEN: `Module`

Technical Breakdown & Key Insight: Must be capitalized: Module. Inheriting from nn.Module registers sub-modules and parameters, enabling automatic device movement.

QUESTION 5 • PyTorch Forward Method **16 Points**

In PyTorch, which core method must be implemented inside a custom nn.Module subclass to define the forward computation logic of the network?

```
class CustomClassifier(nn.Module):
    def __init__(self):
        super().__init__()
        self.fc = nn.Linear(128, 10)

    # Define the forward computational pass:
    def ____(self, x):
        return self.fc(x)
```

CORRECT CODE TOKEN: `forward`

Technical Breakdown & Key Insight: The method is named forward (lowercase). In MindSpore, the equivalent method is named construct — a frequent exam distinction!

QUESTION 6 • PyTorch Inference Mode **16 Points**

During model evaluation and testing in PyTorch, which context manager is used to temporarily deactivate the autograd engine and save memory overhead?

```
model.eval()

# Deactivate autograd tracking during inference:
with torch.____():
    for images, labels in test_loader:
        outputs = model(images)
```

CORRECT CODE TOKEN: `no_grad (or no_grad())`

Technical Breakdown & Key Insight: with torch.no_grad(): disables gradient tracking entirely, speeding up inference and preventing out-of-memory GPU crashes.

QUESTION 7 • PyTorch Evaluation Mode **16 Points**

In PyTorch, which method sets the neural network into evaluation mode, disabling layers such as Dropout and freezing Batch Normalization running statistics?

```
# Switch model from training mode to evaluation mode:
model.____()

with torch.no_grad():
    predictions = model(test_data)
```

CORRECT CODE TOKEN: `eval (or eval())`

Technical Breakdown & Key Insight: model.eval() does NOT disable gradients; it modifies Dropout (disabled) and BatchNorm (uses population running stats).

QUESTION 8 • PyTorch Training Mode **16 Points**

In PyTorch, which method sets the neural network back into training mode before starting an epoch of model optimization?

```
# Switch model into active training mode:
model.____()

for inputs, targets in train_loader:
    optimizer.zero_grad()
```

CORRECT CODE TOKEN: `train (or train())`

Technical Breakdown & Key Insight: model.train() re-enables stochastic dropout layers and batch normalization updates for the next training pass.

QUESTION 9 • PyTorch Dataset Indexing **16 Points**

In PyTorch, when building a custom Dataset by subclassing torch.utils.data.Dataset, which magic method must be overridden to support indexing and fetch a sample by index?

```
from torch.utils.data import Dataset

class ImageDataset(Dataset):
    def __init__(self, data, labels):
        self.data, self.labels = data, labels

    # Fetch sample at specified index:
    def ____(self, idx):
        return self.data[idx], self.labels[idx]
```

CORRECT CODE TOKEN: `__getitem__`

Technical Breakdown & Key Insight: Requires double underscores on both sides: __getitem__. Enables dataset[idx] syntax for DataLoader workers.

QUESTION 10 • PyTorch Dataset Length **16 Points**

In PyTorch, which magic method must be implemented inside a custom Dataset class to return the total size (number of samples) in the dataset?

```
class CustomDataset(Dataset):
    def __init__(self, data):
        self.data = data

    # Return total number of samples:
    def ____(self):
        return len(self.data)
```

CORRECT CODE TOKEN: `__len__`

Technical Breakdown & Key Insight: Requires double underscores: __len__. DataLoader uses this to compute the number of batches per training epoch.

QUESTION 11 • PyTorch State Dict (Saving) **16 Points**

In PyTorch, which method returns a Python dictionary containing the entire mapping of learnable parameter tensors (weights and biases) of a model for saving to disk?

```
# Save only the learned model weights:
torch.save(model.____(), "model_weights.pth")
```

CORRECT CODE TOKEN: `state_dict (or state_dict())`

Technical Breakdown & Key Insight: Huawei best practice recommends saving only the parameters via model.state_dict() rather than pickling the full class architecture.

QUESTION 12 • PyTorch Loading State Dict **16 Points**

In PyTorch, which method is used to load a previously saved parameter state dictionary into an instantiated model architecture?

```
model = LeNet()
weights = torch.load("model_weights.pth")

# Load weights dictionary into model:
model.____(weights)
```

CORRECT CODE TOKEN: `load_state_dict`

Technical Breakdown & Key Insight: Remember the underscores: load_state_dict. It restores all learned weights into the instantiated network object.

QUESTION 13 • PyTorch Classification Loss **16 Points**

Which standard PyTorch loss function in torch.nn is used for multi-class classification, implicitly combining LogSoftmax and Negative Log-Likelihood (NLLLoss)?

```
import torch.nn as nn

# Standard loss function for multi-class classification:
criterion = nn.____()
```

CORRECT CODE TOKEN: `CrossEntropyLoss (or CrossEntropyLoss())`

Technical Breakdown & Key Insight: Do NOT add Softmax to your model output when using nn.CrossEntropyLoss; it already computes LogSoftmax internally.

QUESTION 14 • PyTorch DataLoader Class **16 Points**

Which class from the torch.utils.data module is used to combine a dataset with batching, shuffling, and multi-threaded data loading?

```
from torch.utils.data import Dataset, ____

loader = DataLoader(custom_dataset, batch_size=32, shuffle=True)
```

CORRECT CODE TOKEN: `DataLoader`

Technical Breakdown & Key Insight: PascalCase: DataLoader. It handles batching, memory pinning, shuffling, and multi-worker background loading.

QUESTION 15 • PyTorch Tensor Scalar Extraction **16 Points**

In PyTorch, which method is used to extract a standard Python floating-point scalar value from a 1-element loss tensor for logging?

```
loss = criterion(outputs, labels)

# Convert single-element tensor to python float:
total_loss += loss.____()
```

CORRECT CODE TOKEN: `item (or item())`

Technical Breakdown & Key Insight: Always use loss.item() when logging loss! Calling total_loss += loss retains computation graphs, leaking memory.

Category 2: Huawei MindSpore Framework & Ascend Coding (Q16 – Q25)

QUESTION 16 • MindSpore Cell Base Class **16 Points**

In Huawei's MindSpore framework, which base class must all neural network models and functional layers inherit from (equivalent to PyTorch's nn.Module)?

```
import mindspore.nn as nn

# Base class for all neural network structures in MindSpore:
class LeNet5(nn.____):
    def __init__(self):
        super(LeNet5, self).__init__()
        self.conv1 = nn.Conv2d(1, 6, 5)
```

CORRECT CODE TOKEN: `Cell`

Technical Breakdown & Key Insight: In MindSpore, the foundational class is nn.Cell (capital C). In PyTorch, it is nn.Module.

QUESTION 17 • MindSpore Construct Method **16 Points**

In Huawei MindSpore, which method must be implemented inside a custom nn.Cell to define the forward computational process (equivalent to PyTorch's forward method)?

```
class Network(nn.Cell):
    def __init__(self):
        super().__init__()
        self.dense = nn.Dense(10, 2)

    # MindSpore forward computation method:
    def ____(self, x):
        return self.dense(x)
```

CORRECT CODE TOKEN: `construct`

Technical Breakdown & Key Insight: Top Huawei Exam Trap: MindSpore uses construct, NOT forward! Both take (self, x).

QUESTION 18 • MindSpore High-Level Model API **16 Points**

In MindSpore, which high-level class encapsulates a network, a loss function, and an optimizer to provide high-level training and evaluation interfaces?

```
import mindspore as ms

# Encapsulate training pipeline components:
model = ms.train.____(network, loss_fn, optimizer, metrics={'acc'})
```

CORRECT CODE TOKEN: `Model`

Technical Breakdown & Key Insight: Capitalized: Model. Encapsulates training and inference loops into simple model.train() and model.eval() calls.

QUESTION 19 • MindSpore Model Train Execution **16 Points**

In MindSpore, which method of the Model class is invoked to start model training over a specified number of epochs?

```
model = ms.train.Model(network, loss_fn, optimizer)

# Execute model training:
model.____(epoch=10, train_dataset=ds_train)
```

CORRECT CODE TOKEN: `train`

Technical Breakdown & Key Insight: model.train(epoch, dataset) automatically handles iteration loops, gradient calculation, and metrics monitoring.

QUESTION 20 • MindSpore Model Eval Execution **16 Points**

In MindSpore, which method of the Model class evaluates the performance of the trained network against a validation/test dataset?

```
# Evaluate model metrics on evaluation dataset:
results = model.____(valid_dataset)
print(results)
```

CORRECT CODE TOKEN: `eval`

Technical Breakdown & Key Insight: model.eval(dataset) performs inference on the evaluation dataset and returns the registered dictionary of metric scores.

QUESTION 21 • MindSpore Saving Checkpoint **16 Points**

In MindSpore, which function is used to save the network's learned parameters into a checkpoint file on disk?

```
import mindspore as ms

# Save network parameters to checkpoint file:
ms.____(network, "model_checkpoint.ckpt")
```

CORRECT CODE TOKEN: `save_checkpoint`

Technical Breakdown & Key Insight: Function name: ms.save_checkpoint(network, path). MindSpore checkpoint files traditionally use the .ckpt extension.

QUESTION 22 • MindSpore Loading Checkpoint **16 Points**

In MindSpore, which function loads a parameter dictionary from a saved .ckpt checkpoint file?

```
import mindspore as ms

# Load parameter dictionary from disk:
param_dict = ms.____("model_checkpoint.ckpt")
```

CORRECT CODE TOKEN: `load_checkpoint`

Technical Breakdown & Key Insight: ms.load_checkpoint(path) parses the .ckpt binary and returns a Python dictionary of parameter names and tensor values.

QUESTION 23 • MindSpore Load Param Into Net **16 Points**

In MindSpore, which function takes a loaded parameter dictionary and loads the weights into the neural network instance?

```
param_dict = ms.load_checkpoint("model.ckpt")

# Load parameter dictionary into model instance:
ms.____(network, param_dict)
```

CORRECT CODE TOKEN: `load_param_into_net`

Technical Breakdown & Key Insight: Full function name: ms.load_param_into_net(network, param_dict). Restores trained weights into the network instance.

QUESTION 24 • MindSpore Device Target Context **16 Points**

In MindSpore, to run model training on Huawei's dedicated AI processors, what string must be assigned to device_target in set_context?

```
import mindspore as ms

# Target Huawei dedicated AI processors:
ms.set_context(device_target="____")
```

CORRECT CODE TOKEN: `Ascend`

Technical Breakdown & Key Insight: Must be capitalized: Ascend. Options for device_target include 'Ascend', 'GPU', and 'CPU'.

QUESTION 25 • MindSpore Tensor Class **16 Points**

In MindSpore, what is the core multi-dimensional data structure class used to hold model weights and input arrays?

```
import mindspore as ms
import numpy as np

data = np.array([1.0, 2.0, 3.0], dtype=np.float32)

# Create a MindSpore Tensor:
ms_tensor = ms.____(data, dtype=ms.float32)
```

CORRECT CODE TOKEN: `Tensor`

Technical Breakdown & Key Insight: Capitalized: Tensor. MindSpore Tensors can be initialized directly from Python lists, NumPy arrays, or scalar values.

Category 3: Scikit-Learn Traditional Machine Learning Coding (Q26 – Q35)

QUESTION 26 • Scikit-Learn Model Fitting **16 Points**

In scikit-learn, which core method is called on an estimator instance to train the machine learning algorithm on feature matrix X and target vector y?

```
from sklearn.linear_model import LinearRegression

lr = LinearRegression()

# Train the model on training samples:
lr.____(X_train, y_train)
```

CORRECT CODE TOKEN: `fit`

Technical Breakdown & Key Insight: Universal across scikit-learn: fit(X, y) executes model learning for both supervised and unsupervised estimators.

QUESTION 27 • Scikit-Learn Model Prediction **16 Points**

In scikit-learn, which method is used after training to generate predicted target labels or values for new unseen feature samples?

```
lr.fit(X_train, y_train)

# Predict target values for new test data:
predictions = lr.____(X_test)
```

CORRECT CODE TOKEN: `predict`

Technical Breakdown & Key Insight: Universal across scikit-learn: predict(X_test) returns class predictions for classifiers or scalar values for regressors.

QUESTION 28 • Scikit-Learn Probability Prediction **16 Points**

In scikit-learn classification models (such as LogisticRegression), which method returns the calibrated class probability estimates for each sample?

```
from sklearn.linear_model import LogisticRegression
clf = LogisticRegression()
clf.fit(X_train, y_train)

# Predict class probability distributions:
probabilities = clf.____(X_test)
```

CORRECT CODE TOKEN: `predict_proba`

Technical Breakdown & Key Insight: Notice the underscore: predict_proba. Returns an array of shape *(n_samples, n_classes)* with values between 0 and 1.

QUESTION 29 • Scikit-Learn Fit Transform **16 Points**

In scikit-learn preprocessing (e.g., StandardScaler), which combined method calculates transformation statistics on training data and applies scaling in a single step?

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()

# Compute mean and standard deviation, then scale data:
X_train_scaled = scaler.____(X_train)
```

CORRECT CODE TOKEN: `fit_transform`

Technical Breakdown & Key Insight: fit_transform is strictly for training data: it computes mean/variance AND transforms the feature matrix simultaneously.

QUESTION 30 • Scikit-Learn Transform Only **16 Points**

In scikit-learn preprocessing, which method scales testing data using the previously computed training set statistics without re-estimating parameters?

```
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)

# Apply learned scaling parameters to test set:
X_test_scaled = scaler.____(X_test)
```

CORRECT CODE TOKEN: `transform`

Technical Breakdown & Key Insight: Critical Data Leakage Rule: Never use fit_transform on test data! Always use scaler.transform(X_test).

QUESTION 31 • Scikit-Learn Train Test Split **16 Points**

In scikit-learn, which function from model_selection is commonly used to randomly partition a dataset into training and testing subsets?

```
from sklearn.model_selection import ____

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

CORRECT CODE TOKEN: `train_test_split`

Technical Breakdown & Key Insight: Imported from sklearn.model_selection: train_test_split randomly divides data according to test_size.

QUESTION 32 • Linear Regression Coef Attribute **16 Points**

In scikit-learn's LinearRegression, which attribute stores the learned weight coefficients (slopes) of the model after fitting?

```
lr = LinearRegression()
lr.fit(X, y)

# Access learned weight parameters:
weights = lr.____
```

CORRECT CODE TOKEN: `coef_`

Technical Breakdown & Key Insight: Trailing underscore convention: coef_. In scikit-learn, all attributes learned during fit() end with an underscore.

QUESTION 33 • Linear Regression Intercept Attribute **16 Points**

In scikit-learn's LinearRegression, which attribute stores the learned bias (intercept) of the model after fitting?

```
lr = LinearRegression()
lr.fit(X, y)

# Access learned intercept/bias parameter:
bias = lr.____
```

CORRECT CODE TOKEN: `intercept_`

Technical Breakdown & Key Insight: Trailing underscore convention: intercept_. Represents the constant bias term (the y-intercept where X=0).

QUESTION 34 • K-Means Cluster Count Parameter **16 Points**

In scikit-learn's KMeans clustering estimator, which hyperparameter specifies the number of clusters to form?

```
from sklearn.cluster import KMeans

# Initialize KMeans with 3 target clusters:
kmeans = KMeans(____=3, random_state=42)
```

CORRECT CODE TOKEN: `n_clusters`

Technical Breakdown & Key Insight: Hyperparameter name: n_clusters (not k, not num_clusters). Corresponds to the parameter K in K-Means clustering.

QUESTION 35 • K-Means Cluster Centroids Attribute **16 Points**

In scikit-learn's KMeans estimator, which attribute stores the final coordinates of cluster centers after training?

```
kmeans = KMeans(n_clusters=3)
kmeans.fit(X)

# Retrieve learned cluster centers:
centers = kmeans.____
```

CORRECT CODE TOKEN: `cluster_centers_`

Technical Breakdown & Key Insight: Trailing underscore: cluster_centers_. Returns an array of coordinates for the K learned cluster centers.

Category 4: NumPy & Tensor Preprocessing Foundations (Q36 – Q40)

QUESTION 36 • NumPy Zeros Creation **16 Points**

In NumPy, which function initializes an array of given shape and type populated entirely with zeros?

```
import numpy as np

# Create a 3x3 array filled with zeros:
zero_matrix = np.____((3, 3))
```

CORRECT CODE TOKEN: `zeros`

Technical Breakdown & Key Insight: np.zeros((shape)) creates an array of zeros. Useful for allocating accumulator buffers and weight matrices.

QUESTION 37 • NumPy Ones Creation **16 Points**

In NumPy, which function initializes an array of given shape and type populated entirely with ones?

```
import numpy as np

# Create an array of shape (4, 2) filled with ones:
ones_matrix = np.____((4, 2))
```

CORRECT CODE TOKEN: `ones`

Technical Breakdown & Key Insight: np.ones((shape)) creates an array populated with ones. Frequently used to initialize bias vectors or uniform masks.

QUESTION 38 • NumPy Reshape Function **16 Points**

In NumPy, which method gives a new shape to an array without changing its underlying data elements?

```
arr = np.arange(10)  # Shape (10,)

# Reshape 1D array into 2D matrix of 2 rows and 5 columns:
matrix = arr.____((2, 5))
```

CORRECT CODE TOKEN: `reshape`

Technical Breakdown & Key Insight: arr.reshape(new_shape) changes the tensor dimensions while keeping the total number of elements identical.

QUESTION 39 • NumPy Array Shape Attribute **16 Points**

In NumPy, which attribute of an ndarray returns a tuple representing the dimensions of the array?

```
arr = np.zeros((3, 4, 5))

# Get dimensions of array:
dims = arr.____  # Returns (3, 4, 5)
```

CORRECT CODE TOKEN: `shape`

Technical Breakdown & Key Insight: arr.shape is an attribute (no parentheses!). Returns a tuple like (batch_size, channels, height, width).

QUESTION 40 • NumPy Type Casting **16 Points**

In NumPy, which method casts an array to a specified numerical data type (such as np.float32)?

```
int_arr = np.array([1, 2, 3])

# Cast array to 32-bit floating point type:
float_arr = int_arr.____(np.float32)
```

CORRECT CODE TOKEN: `astype`

Technical Breakdown & Key Insight: arr.astype(dtype) creates a copy of the array cast to the specified type, essential when preparing inputs for deep learning models.

HCIA-AI V4.0 Code-Writing Blanks: Quick Reference Summary

Complete alphabetical review table of all 40 essential code completion tokens across PyTorch, MindSpore, Scikit-Learn, and NumPy.

Q#	Category	Target Code Context	Accepted Code Token
Q1	PyTorch Autograd	In the PyTorch training loop, which method must be called on the ...	zero_grad (or zero_grad())
Q2	PyTorch Backpropagation	In PyTorch, which method is invoked on the computed scalar loss t...	backward (or backward())
Q3	PyTorch Optimizer Update	In PyTorch, which method of the optimizer is called to update the...	step (or step())
Q4	PyTorch Model Inheritance	In PyTorch, all user-defined neural network architectures must in...	Module
Q5	PyTorch Forward Method	In PyTorch, which core method must be implemented inside a custom...	forward
Q6	PyTorch Inference Mode	During model evaluation and testing in PyTorch, which context man...	no_grad (or no_grad())
Q7	PyTorch Evaluation Mode	In PyTorch, which method sets the neural network into evaluation ...	eval (or eval())
Q8	PyTorch Training Mode	In PyTorch, which method sets the neural network back into traini...	train (or train())
Q9	PyTorch Dataset Indexing	In PyTorch, when building a custom Dataset by subclassing torch.u...	__getitem__
Q10	PyTorch Dataset Length	In PyTorch, which magic method must be implemented inside a custo...	__len__
Q11	PyTorch State Dict (Saving)	In PyTorch, which method returns a Python dictionary containing t...	state_dict (or state_dict())
Q12	PyTorch Loading State Dict	In PyTorch, which method is used to load a previously saved param...	load_state_dict
Q13	PyTorch Classification Loss	Which standard PyTorch loss function in torch.nn is used for mult...	CrossEntropyLoss (or CrossEntropyLoss())
Q14	PyTorch DataLoader Class	Which class from the torch.utils.data module is used to combine a...	DataLoader
Q15	PyTorch Tensor Scalar Extraction	In PyTorch, which method is used to extract a standard Python flo...	item (or item())
Q16	MindSpore Cell Base Class	In Huawei's MindSpore framework, which base class must all neural...	Cell
Q17	MindSpore Construct Method	In Huawei MindSpore, which method must be implemented inside a cu...	construct
Q18	MindSpore High-Level Model API	In MindSpore, which high-level class encapsulates a network, a lo...	Model
Q19	MindSpore Model Train Execution	In MindSpore, which method of the Model class is invoked to start...	train
Q20	MindSpore Model Eval Execution	In MindSpore, which method of the Model class evaluates the perfo...	eval
Q21	MindSpore Saving Checkpoint	In MindSpore, which function is used to save the network's learne...	save_checkpoint

Q22	MindSpore Loading Checkpoint	In MindSpore, which function loads a parameter dictionary from a ...	**load_checkpoint**
Q23	MindSpore Load Param Into Net	In MindSpore, which function takes a loaded parameter dictionary ...	**load_param_into_net**
Q24	MindSpore Device Target Context	In MindSpore, to run model training on Huawei's dedicated AI proc...	**Ascend**
Q25	MindSpore Tensor Class	In MindSpore, what is the core multi-dimensional data structure c...	**Tensor**
Q26	Scikit-Learn Model Fitting	In scikit-learn, which core method is called on an estimator inst...	**fit**
Q27	Scikit-Learn Model Prediction	In scikit-learn, which method is used after training to generate ...	**predict**
Q28	Scikit-Learn Probability Prediction	In scikit-learn classification models (such as LogisticRegression...	**predict_proba**
Q29	Scikit-Learn Fit Transform	In scikit-learn preprocessing (e.g., StandardScaler), which combi...	**fit_transform**
Q30	Scikit-Learn Transform Only	In scikit-learn preprocessing, which method scales testing data u...	**transform**
Q31	Scikit-Learn Train Test Split	In scikit-learn, which function from model_selection is commonly ...	**train_test_split**
Q32	Linear Regression Coef Attribute	In scikit-learn's LinearRegression, which attribute stores the le...	**coef_**
Q33	Linear Regression Intercept Attribute	In scikit-learn's LinearRegression, which attribute stores the le...	**intercept_**
Q34	K-Means Cluster Count Parameter	In scikit-learn's KMeans clustering estimator, which hyperparamet...	**n_clusters**
Q35	K-Means Cluster Centroids Attribute	In scikit-learn's KMeans estimator, which attribute stores the fi...	**cluster_centers_**
Q36	NumPy Zeros Creation	In NumPy, which function initializes an array of given shape and ...	**zeros**
Q37	NumPy Ones Creation	In NumPy, which function initializes an array of given shape and ...	**ones**
Q38	NumPy Reshape Function	In NumPy, which method gives a new shape to an array without chan...	**reshape**
Q39	NumPy Array Shape Attribute	In NumPy, which attribute of an ndarray returns a tuple represent...	**shape**
Q40	NumPy Type Casting	In NumPy, which method casts an array to a specified numerical da...	**astype**